

Exhaustive verification of Seymour’s Second Neighborhood Conjecture for oriented graphs on at most 14 vertices*

Automated verification pipeline (Atris workspace)

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Abstract

Seymour’s Second Neighborhood Conjecture states that every oriented graph (finite simple digraph with no loops and no 2-cycles) contains a vertex v with $|N^{++}(v)| \geq |N^+(v)|$, where $N^{++}(v)$ is the set of vertices at directed distance exactly two from v . We report an exhaustive computational verification: no counterexample exists on $n \leq 14$ vertices. The result rests on 33 unsatisfiable SAT instances solved by CaDiCaL 3.0.1 with DRAT proof logging, every proof checked by drat-trim; the instances cover the search space through a minimum out-degree sharding scheme whose covering and symmetry-breaking arguments are short pen-and-paper proofs reproduced here. (The full artifact set spans $n \leq 18$ and 63 CNF instances, of which 59 refutations are complete; the 33 form the $n \leq 14$ claim.) Independently of the SAT chain, all oriented graphs on $n \leq 8$ vertices (575,016,219 at $n = 8$) were enumerated with nauty and checked by a from-scratch C program, with per-order counts matching OEIS A001174 exactly and zero counterexamples. The CaDiCaL runs also close every shard for $n = 15$ and $n = 16$; at the time of writing the DRAT proofs of the two largest shards (0.6 GB and 4.2 GB) had not finished re-verification, so we state $n \leq 14$ as the fully machine-checked claim and report $n = 15, 16$ separately with their exact status. We found no previously published exhaustive verification of the conjecture over all oriented graphs of a given order, so we state the small-order record here explicitly rather than compare against one.

1 The conjecture and its history

Let D be an *oriented graph*: a finite directed graph with no loops, no parallel arcs, and no digons (if (u, v) is an arc then (v, u) is not). For a vertex v write $N^+(v)$ for its out-neighborhood and $N^{++}(v)$ for its *second out-neighborhood*, the set of vertices at directed distance exactly 2 from v . A vertex v with $|N^{++}(v)| \geq |N^+(v)|$ is called a *Seymour vertex*.

Conjecture 1 (Seymour, c. 1990; see Dean and Latka [1]). *Every oriented graph has a Seymour vertex.*

A counterexample is an oriented graph in which *every* vertex v satisfies $|N^{++}(v)| < |N^+(v)|$. The conjecture is open. The main published partial results are:

*Working note, not peer reviewed. The computations and this text were produced by an automated agent pipeline (Claude Fable 5, Atris workspace of Keshav Rao) on a single shared Apple Silicon laptop, 2026-07-22/23. Section 6 lists exactly what a human reviewer must audit before any of this is submitted anywhere. No claim in this note should be trusted beyond the evidence described in it.

- **Tournaments.** Dean’s conjecture (the tournament case) was proved by Fisher [2] using an averaging argument over probability distributions; Havet and Thomassé [3] gave a combinatorial proof via median orders, producing two Seymour vertices when the tournament has no dominated vertex.
- **Small minimum out-degree.** Kaneko and Locke [4] proved the conjecture for oriented graphs with minimum out-degree at most 6. A June 2026 preprint by Sadhukhan, Sandeep, and Sen [10] claims the case of minimum out-degree exactly 7, using local reductions plus computer-assisted infeasibility checks; it has not, to our knowledge, been refereed.
- **Approximate versions.** Chen, Shen, and Yuster [5] proved every oriented graph has a vertex with $|N^{++}(v)| \geq \gamma |N^+(v)|$ for $\gamma = 0.657\dots$, with later improvements to the constant by others.
- **Random and dense settings.** Cohn, Godbole, Wright Harkness, and Zhang [8] showed the conjecture holds asymptotically almost surely for random tournaments and random digraphs. Espuny Díaz, Girão, Granet, and Kronenberg [9] proved that a.a.s. every orientation of $G(n, p)$ satisfies the conjecture for fixed $p < 1/2$, and gave two reductions; their Proposition 4 shows a vertex-minimal counterexample D must have minimum out-degree greater than $\sqrt{|V(D)|}$.
- **Structural classes.** Fidler and Yuster [6] handled tournaments minus a star or a subtournament and digraphs of minimum degree $|V(D)| - 2$; Brantner, Brockman, Kay, and Snively [7] handled digraphs whose underlying graph is triangle-free.

On the exhaustive small-order record. We searched for a published exhaustive verification of Conjecture 1 over *all* oriented graphs on $n \leq k$ vertices for any k and found none: neither the standard open-problem surveys [11, 12] nor the papers above report one. An earlier internal working note of ours asserted a published record of $n \leq 7$; we could not trace that assertion to any source and we withdraw it. The correct statement appears to be that no such record was published, presumably because checks up to $n \approx 8$ are routine with nauty. The contribution of this note is therefore not the beating of a record but the explicit, certificate-backed statement of one.

Theorem 1 (main claim, machine-checked). *No oriented graph on at most 14 vertices is a counterexample to Conjecture 1.*

Combined with Kaneko and Locke [4] (any counterexample has minimum out-degree at least 7) and the arc-counting bound of Section 2.2 (minimum out-degree at most $\lfloor (n-1)/2 \rfloor$), a hypothetical counterexample must in fact have $n \geq 15$; Theorem 1 is the unconditional statement that does not rely on [4], and our shards re-prove the low-degree cases for $n \leq 14$ from scratch rather than citing them.

2 Method

The overall shape: for each order n , we encode “an oriented graph on n labeled vertices in which every vertex violates the Seymour inequality” as a CNF formula, split the space into minimum out-degree shards with symmetry breaking, obtain UNSAT verdicts with DRAT proofs from

CaDiCaL 3.0.1 [13], and check every proof with drat-trim [14]. The DRAT certificates prove the CNFs unsatisfiable; the link from the CNFs to the conjecture is the soundness chain below, which is deliberately short and must be audited by a human.

2.1 Encoding

The encoder is `encode.py` (about 150 lines, PySAT [15] 1.9). Variables: x_{uv} for each ordered pair ($u \rightarrow v$ arc), and y_{vw} for each ordered pair with $v \neq w$, an *upper indicator* for $w \in N^{++}(v)$. Clauses:

1. $(\neg x_{uv} \vee \neg x_{vu})$ for all $u < v$: no digons. No variable x_{vv} exists, so loops cannot be represented.
2. $(x_{vu} \wedge x_{uw} \wedge \neg x_{vw}) \rightarrow y_{vw}$ for all distinct u, v, w : every genuine 2-path not shortcut by an arc forces y_{vw} up. Hence in every model, $\sum_w y_{vw} \geq |N^{++}(v)|$.
3. For each v : a totalizer cardinality constraint [16] encoding $\sum_w y_{vw} + \sum_u \neg x_{vu} \leq n - 2$, which given the previous point is equivalent to $|N^{++}(v)| \leq |N^+(v)| - 1$, i.e., v violates the Seymour inequality.

Completeness (the direction that matters for a nonexistence claim): if a counterexample D on n labeled vertices exists, set x from the arcs of D and set y_{vw} to the exact indicator of $w \in N^{++}(v)$. Group (1) holds because D is an oriented graph; group (2) holds because any 2-path $v \rightarrow u \rightarrow w$ with no arc $v \rightarrow w$ witnesses $w \in N^{++}(v)$; group (3) holds because every vertex of D violates the inequality. So the CNF is satisfiable, and UNSAT verdicts rule out every labeled counterexample on n vertices. Spurious $y = 1$ assignments only tighten constraint (3), so soundness of the search is not an issue for the nonexistence direction.

For $n \leq 9$ we ran this formula over the whole space with no symmetry breaking and a single additional constraint, minimum out-degree ≥ 1 (the encoder's default): this excludes only graphs containing a sink, and a sink v has $|N^{++}(v)| = 0 = |N^+(v)|$, so it is itself a Seymour vertex and no graph containing one is a counterexample. Beyond that one elementary observation the four instances are assumption-free: their verified refutations alone prove Theorem 1 for $n \leq 9$.

2.2 Degree shards and symmetry breaking

For $n \geq 10$ the whole-space formula becomes slow, so the space is split. In a counterexample no vertex is a sink (a sink v has $|N^{++}(v)| = 0 = |N^+(v)|$ and would be a Seymour vertex), so the minimum out-degree d_0 satisfies $d_0 \geq 1$; counting arcs, $n d_0 \leq \sum_v \deg^+(v) = |A(D)| \leq \binom{n}{2}$, so $d_0 \leq \lfloor (n-1)/2 \rfloor$. Shard s_d (for $1 \leq d \leq \lfloor (n-1)/2 \rfloor$) asserts:

- every vertex has out-degree $\geq d$ (totalizers),
- vertex 0 has out-degree exactly d (totalizers),
- $N^+(0) = \{1, \dots, d\}$ (unit clauses),
- two-block lex-leader symmetry breaking: for every adjacent transposition σ acting within $\{1, \dots, d\}$ or within $\{d+1, \dots, n-1\}$, the adjacency matrix A satisfies $A \leq_{\text{lex}} A \circ \sigma$ [17].

Covering argument. Suppose a counterexample D on n vertices exists and let d_0 be its minimum out-degree, so $1 \leq d_0 \leq \lfloor (n-1)/2 \rfloor$. Relabel a minimum out-degree vertex to 0 and its out-neighbors to $1, \dots, d_0$. The residual relabeling freedom is exactly the subgroup $H = \text{Sym}\{1, \dots, d_0\} \times \text{Sym}\{d_0 + 1, \dots, n-1\}$, which preserves the pinned constraints. Choose the labeling whose adjacency matrix is lexicographically minimal over the H -orbit; it satisfies $A \leq_{\text{lex}} A \circ \sigma$ for every $\sigma \in H$, in particular for the encoded adjacent transpositions. Hence some assignment satisfies shard s_{d_0} , and UNSAT verdicts on all shards of order n rule out all counterexamples on n vertices. Note the encoded constraints are a *subset* (adjacent transpositions only) of the full lex-leader condition; using a subset weakens the formula and therefore cannot cause a false UNSAT.

For $n = 10$ we additionally ran the whole space in one piece (minimum out-degree ≥ 1 , full-lex symmetry breaking over all adjacent transpositions of the symmetric group, which is sound because the formula without pinning is invariant under all vertex permutations), so Theorem 1 at $n = 10$ does not depend on the shard argument either. Cross-checks at $n = 10, 11$ with the (audited, elementary) bound “any counterexample has minimum out-degree ≥ 3 ” agree; that lemma is stated in the artifact repository but is not load-bearing anywhere in this note, precisely because shards s_1 and s_2 were machine-proved directly.

2.3 Tournament-forced shards and Lemma T

For odd n , the top shard $d = (n-1)/2$ forces $\sum_v \deg^+(v) \geq n(n-1)/2$, i.e., every pair adjacent: the graph is a $\frac{n-1}{2}$ -regular tournament. These shards are closed twice over:

1. By Fisher’s theorem [2]: every tournament has a Seymour vertex, so no tournament is a counterexample. This is published, peer-reviewed mathematics and is the citation we prefer.
2. By the following elementary argument, produced in-session and **not yet reviewed by any human**; we record it because it is short enough to audit in one sitting.

Lemma 1 (Lemma T, **unreviewed**). *In a k -regular tournament, every vertex v satisfies $|N^{++}(v)| = |N^+(v)| = k$, so no such tournament is a counterexample.*

Proof (to be audited). Fix v and take any $w \notin \{v\} \cup N^+(v)$. If $w \notin N^{++}(v)$, then no $u \in N^+(v)$ has $u \rightarrow w$; by totality of the tournament, $w \rightarrow u$ for all $u \in N^+(v)$, and also $w \rightarrow v$ (since $v \rightarrow w$ would put w in $N^+(v)$). Then $\deg^+(w) \geq k+1$, contradicting k -regularity. Hence $N^{++}(v) = V \setminus (\{v\} \cup N^+(v))$, which has size $n-1-k = k$ since $n = 2k+1$. $\square$

For $n \leq 14$ neither item is needed (no tournament-forced shard occurs with an unfinished SAT run there). They matter only for the $n = 15$ and $n = 17$ top shards discussed in Section 5.2; and for $n = 15$ the SAT run of the tournament shard itself finished UNSAT as well, so Lemma T is a third line of defense there, not a dependency.

3 Two independent pipelines

3.1 Pipeline A: SAT with DRAT certificates

Every instance in Table 1 was solved by CaDiCaL 3.0.1 [13] with binary DRAT proof output. Every completed proof was checked with drat-trim [14] built from source (commit 2e3b2dc,

upstream `marijnheule/drat-trim`). Of the 59 completed refutations, 57 have `s VERIFIED` receipts at freeze time; the remaining two are the giant `n15_s7` and `n16_s7` proofs whose checks were still running (see `receipts/SUMMARY.txt` for the final word; every receipt generated so far reports `s VERIFIED` and none has ever failed). The receipts were *regenerated from scratch in the write-up session* (2026-07-23) by a different agent than the one that ran the solver, and are stored alongside this note (`receipts/`). As a negative control, a truncated copy of one proof was rejected by `drat-trim` (`s NOT VERIFIED`), confirming the checker does not rubber-stamp.

Additionally, all 63 CNF files on disk were regenerated byte-identically from `encode.py` with their documented flags in the write-up session, tying the artifacts that were solved to the encoder that was audited: the four whole-space files ($n = 6, \dots, 9$) with no flags, the three cross-check files with `--mindeg` and full-lex flags, and all 56 shard files with `--mindeg d --exactdeg0 d --pin-nbhd --sb-mode twoblock`.

3.2 Pipeline B: exhaustive enumeration, $n \leq 8$

Independently of SAT, encoder, and solver: nauty’s [18] `geng` generated all graphs, `directg -o` all their orientations, and a from-scratch C checker (`check_d6.c`, about 80 lines, reading `digraph6` directly and re-checking the no-loop/no-digon property itself) tested every oriented graph on $n \leq 8$ vertices. Counts match OEIS A001174 [19] (number of oriented graphs on n unlabeled vertices) exactly at every order, which is strong evidence the enumeration is complete:

n	1	2	3	4	5	6	7	8
oriented graphs	1	2	7	42	582	21,480	2,142,288	575,016,219
counterexamples	0	0	0	0	0	0	0	0

The $n \leq 7$ runs were re-executed live during the write-up session (seconds); the $n = 8$ run was sharded ten ways (`geng`’s `res/mod` splitting) and the per-shard `checked` counts sum to exactly 575,016,219. A positive control (`check_weak`, the same checker with the comparison deliberately weakened) flags all 7 oriented graphs on 3 vertices, as it must.

So for $n \leq 8$ the result is double-proved by disjoint toolchains (PySAT/CaDiCaL/drat-trim versus nauty/hand-written C), and for $n = 9, \dots, 14$ it rests on Pipeline A plus the written arguments of Section 2.

3.3 Auxiliary lane: Cayley digraphs of small groups

As a targeted probe beyond $n = 14$ (not part of the headline claim), all 55 groups of orders 12 through 24 were constructed and brute-force verified to be groups, with isomorphism-type counts per order matching the classification (5, 1, 2, 1, 14, 1, 5, 1, 5, 2, 2, 1, 15); for each group, every connection set S with $S \cap S^{-1} = \emptyset$ was enumerated (1,505,160 sets total) and the resulting Cayley digraph checked at the identity vertex, vertex-transitivity justifying the single-vertex check; that shortcut was itself revalidated by full all-vertex checks on every 250th set. Zero counterexamples. This lane shares no code with either pipeline above but has not been independently re-audited; treat it as a search result, not a certified one.

4 Controls

Beyond the negative proof-checker control and the byte-identical CNF regeneration already described:

- **Slack controls (SAT expected).** Relaxing the per-vertex bound by one (encoding $|N^{++}(v)| \leq |N^+(v)|$) must be satisfiable, since, e.g., directed cycles and their blowups sit exactly on the boundary. This was confirmed at $n = 3$ (plain) and, critically, at $n = 6$ and $n = 12$ *through the full shard machinery* (pinning, exact degree, two-block lex): the shard path is not accidentally over-constrained. The $n = 12$ control model was decoded back into an explicit digraph and checked to be a genuine boundary witness, validating the totalizer calibration end to end.
- **Shard UNSAT control.** A shard instance at $n = 7$, where Pipeline B independently proves no counterexample exists, is UNSAT as expected. All four control instances were re-solved from the stored CNF files during the write-up session with the expected verdicts (SAT, SAT, UNSAT, SAT).
- **Solver agreement.** Spot-checked shards were re-solved with Glucose 4.2 and MiniSat 2.2 with agreeing verdicts (no proofs logged from those; CaDiCaL’s DRAT proofs are the receipts).

5 Results

Table 1 lists every SAT instance with its DIMACS size, CaDiCaL wall time, DRAT proof size, verdict, and this session’s drat-trim receipt. The table is generated mechanically from the artifacts by `gen_tables.py`; no number in it was typed by hand. Wall times are from a heavily shared machine and should be read as upper bounds. Shard counts per order equal $\lfloor (n-1)/2 \rfloor$ exactly: 4 shards at $n = 10$, then 5, 5, 6, 6, 7, 7, 8, 8 through $n = 18$.

Table 1: All SAT instances. Columns: DIMACS size, CaDiCaL wall time (seconds), DRAT proof size, solver verdict, and the drat-trim receipt regenerated in the write-up session (2026-07-23).

instance	meaning	vars	clauses	time	proof	verdict	drat-trim
n6	$n=6$ whole space, $\delta^+ \geq 1$, no SB	264	621	<0.1	–	UNSAT	VERIFIED
n7	$n=7$ whole space, $\delta^+ \geq 1$, no SB	392	1015	0.4	–	UNSAT	VERIFIED
n8	$n=8$ whole space, $\delta^+ \geq 1$, no SB	544	1540	10	–	UNSAT	VERIFIED
n9	$n=9$ whole space, $\delta^+ \geq 1$, no SB	720	2214	279	–	UNSAT	VERIFIED
n10_d1_sb	$n=10$ whole space, $\delta^+ \geq 1$, full lex	1246	4902	27	–	UNSAT	VERIFIED
n10_d3_sb	$n=10$, $\delta^+ \geq 3$ (lemma), full lex	1536	5552	15	–	UNSAT	VERIFIED
n11_d3_sb	$n=11$, $\delta^+ \geq 3$ (lemma), full lex	1942	7264	503	–	UNSAT	VERIFIED
n10_s1	$n=10$, shard $d_0=1$	1207	4571	<0.1	–	UNSAT	VERIFIED
n10_s2	$n=10$, shard $d_0=2$	1497	5221	<0.1	–	UNSAT	VERIFIED
n10_s3	$n=10$, shard $d_0=3$	1497	5221	<0.1	–	UNSAT	VERIFIED
n10_s4	$n=10$, shard $d_0=4$	1497	5221	<0.1	–	UNSAT	VERIFIED
n11_s1	$n=11$, shard $d_0=1$	1526	6031	<0.1	–	UNSAT	VERIFIED
n11_s2	$n=11$, shard $d_0=2$	1900	6900	<0.1	–	UNSAT	VERIFIED
n11_s3	$n=11$, shard $d_0=3$	1900	6900	<0.1	–	UNSAT	VERIFIED
n11_s4	$n=11$, shard $d_0=4$	1900	6900	<0.1	–	UNSAT	VERIFIED
n11_s5	$n=11$, shard $d_0=5$	1900	6900	<0.1	–	UNSAT	VERIFIED

instance	meaning	vars	clauses	time	proof	verdict	drat-trim
n12_s1	$n=12$, shard $d_0=1$	1881	7747	<0.1	–	UNSAT	VERIFIED
n12_s2	$n=12$, shard $d_0=2$	2349	8875	<0.1	–	UNSAT	VERIFIED
n12_s3	$n=12$, shard $d_0=3$	2349	8875	<0.1	–	UNSAT	VERIFIED
n12_s4	$n=12$, shard $d_0=4$	2349	8875	<0.1	–	UNSAT	VERIFIED
n12_s5	$n=12$, shard $d_0=5$	2349	8875	0.2	–	UNSAT	VERIFIED
n13_s1	$n=13$, shard $d_0=1$	2272	9737	<0.1	–	UNSAT	VERIFIED
n13_s2	$n=13$, shard $d_0=2$	2844	11167	<0.1	–	UNSAT	VERIFIED
n13_s3	$n=13$, shard $d_0=3$	2844	11167	<0.1	–	UNSAT	VERIFIED
n13_s4	$n=13$, shard $d_0=4$	2844	11167	<0.1	–	UNSAT	VERIFIED
n13_s5	$n=13$, shard $d_0=5$	2844	11167	0.7	–	UNSAT	VERIFIED
n13_s6	$n=13$, shard $d_0=6$	2844	11167	10.0	–	UNSAT	VERIFIED
n14_s1	$n=14$, shard $d_0=1$	2699	12019	<0.1	–	UNSAT	VERIFIED
n14_s2	$n=14$, shard $d_0=2$	3385	13797	<0.1	–	UNSAT	VERIFIED
n14_s3	$n=14$, shard $d_0=3$	3385	13797	<0.1	–	UNSAT	VERIFIED
n14_s4	$n=14$, shard $d_0=4$	3385	13797	<0.1	–	UNSAT	VERIFIED
n14_s5	$n=14$, shard $d_0=5$	3385	13797	2.3	–	UNSAT	VERIFIED
n14_s6	$n=14$, shard $d_0=6$	3385	13797	41	–	UNSAT	VERIFIED
n15_s1	$n=15$, shard $d_0=1$	3162	14611	<0.1	–	UNSAT	VERIFIED
n15_s2	$n=15$, shard $d_0=2$	3972	16786	<0.1	–	UNSAT	VERIFIED
n15_s3	$n=15$, shard $d_0=3$	3972	16786	<0.1	–	UNSAT	VERIFIED
n15_s4	$n=15$, shard $d_0=4$	3972	16786	<0.1	–	UNSAT	VERIFIED
n15_s5	$n=15$, shard $d_0=5$	3972	16786	2.8	–	UNSAT	VERIFIED
n15_s6	$n=15$, shard $d_0=6$	3972	16786	51	–	UNSAT	VERIFIED
n15_s7	$n=15$, shard $d_0=7$	3972	16786	4643	–	UNSAT	pending
n16_s1	$n=16$, shard $d_0=1$	3661	17531	<0.1	–	UNSAT	VERIFIED
n16_s2	$n=16$, shard $d_0=2$	4605	20155	<0.1	–	UNSAT	VERIFIED
n16_s3	$n=16$, shard $d_0=3$	4605	20155	<0.1	–	UNSAT	VERIFIED
n16_s4	$n=16$, shard $d_0=4$	4605	20155	<0.1	–	UNSAT	VERIFIED
n16_s5	$n=16$, shard $d_0=5$	4605	20155	3.9	–	UNSAT	VERIFIED
n16_s6	$n=16$, shard $d_0=6$	4605	20155	116	–	UNSAT	VERIFIED
n16_s7	$n=16$, shard $d_0=7$	4605	20155	19080	–	UNSAT	pending
n17_s1	$n=17$, shard $d_0=1$	4196	20797	<0.1	–	UNSAT	VERIFIED
n17_s2	$n=17$, shard $d_0=2$	5284	23925	<0.1	–	UNSAT	VERIFIED
n17_s3	$n=17$, shard $d_0=3$	5284	23925	<0.1	–	UNSAT	VERIFIED
n17_s4	$n=17$, shard $d_0=4$	5284	23925	<0.1	–	UNSAT	VERIFIED
n17_s5	$n=17$, shard $d_0=5$	5284	23925	2.6	–	UNSAT	VERIFIED
n17_s6	$n=17$, shard $d_0=6$	5284	23925	84	–	UNSAT	VERIFIED
n17_s7	$n=17$, shard $d_0=7$	5284	23925	–	–	running	–
n17_s8	$n=17$, shard $d_0=8$	5284	23925	–	–	running	–
n18_s1	$n=18$, shard $d_0=1$	4804	24464	<0.1	–	UNSAT	VERIFIED
n18_s2	$n=18$, shard $d_0=2$	6064	28172	<0.1	–	UNSAT	VERIFIED
n18_s3	$n=18$, shard $d_0=3$	6064	28172	<0.1	–	UNSAT	VERIFIED
n18_s4	$n=18$, shard $d_0=4$	6064	28172	<0.1	–	UNSAT	VERIFIED
n18_s5	$n=18$, shard $d_0=5$	6064	28172	2.9	–	UNSAT	VERIFIED
n18_s6	$n=18$, shard $d_0=6$	6064	28172	113	–	UNSAT	VERIFIED
n18_s7	$n=18$, shard $d_0=7$	6064	28172	–	–	running	–
n18_s8	$n=18$, shard $d_0=8$	6064	28172	–	–	running	–

5.1 Fully verified claim

For $n \leq 14$: every listed instance is UNSAT with a drat-trim-verified DRAT proof. By the completeness argument of Section 2.1 (for $n \leq 9$; the $n = 10$ whole-space instance additionally uses the full-lex soundness argument of Section 2.2) and the covering argument of Section 2.2 (for the sharded orders), this proves Theorem 1. For $n \leq 8$, Pipeline B independently confirms

it.

5.2 Beyond 14: exact status at write-up time

- **n = 15:** all 7 shards UNSAT by CaDiCaL (top shard s_7 : 4,643s wall, 0.60 GB proof). Shards s_1 through s_6 have verified DRAT proofs; verification of the s_7 proof was still running when this note was frozen (its receipt, if present in `receipts/`, supersedes this sentence). Independently of that one certificate, shard s_7 forces a 7-regular tournament and is closed by Fisher’s theorem [2], and also by the unreviewed Lemma 1. We therefore regard $n = 15$ as established modulo one of: (a) the pending big-proof verification, or (b) reliance on Fisher [2], and we deliberately keep it out of the headline.
- **n = 16:** all 7 shards UNSAT by CaDiCaL (s_7 : 19,080s wall, 4.23 GB proof); s_1 through s_6 verified; verification of the s_7 proof still running. No tournament argument applies at even n , so $n = 16$ waits on that verification.
- **n = 17, 18:** shards s_1 through s_6 UNSAT and verified. The near-tournament shards ($n17_s7$, $n17_s8$, $n18_s7$, $n18_s8$) were still being solved at write-up time (> 5 hours each). $n17_s8$ forces an 8-regular tournament and is closed by Fisher [2] (or Lemma 1) regardless of its SAT run; $n17_s7$, $n18_s7$, $n18_s8$ are genuinely open runs. So $n = 17, 18$ are *not* claimed.

6 What an external reviewer must audit

This note was produced by an automated pipeline. Before any submission, a human must independently check, at minimum:

1. **The soundness chain (Section 2).** The DRAT certificates prove only that specific CNF files are unsatisfiable. The encoding-completeness argument, the sink and arc-count bounds on d_0 , and the two-block lex-leader covering argument are pen-and-paper claims living outside all certificates. They are short; audit every line.
2. **encode.py against Section 2.1.** About 150 lines. Confirm the clause groups match the stated encoding and that PySAT’s totalizer (`CardEnc.atmost`, `EncType.totalizer`) is used correctly (auxiliary variables must not be shared across constraints incorrectly; the slack controls test this but do not prove it).
3. **Lemma 1**, or simply replace every use of it by the citation to Fisher [2] and delete it.
4. **Re-verification of all certificates on independent hardware**, rebuilding CaDiCaL and drat-trim from upstream sources. The full artifact set is about 28 GB; SHA-256 checksums of every CNF, every completed proof, and the three load-bearing source files are in `receipts/SHA256SUMS.txt`. Ideally re-check with a second, independently implemented DRAT checker (e.g., the formally verified `cake_lpr`).
5. **Pipeline B.** Read `check_d6.c` (about 80 lines, including its own digraph6 parser and digon check); confirm the `geng/directg -o` invocations enumerate exactly the oriented graphs; re-derive the OEIS comparison.

6. **Provenance discipline.** Only `encode.py`, `check_d6.c`, the CNF/DRAT artifacts, and the receipts are load-bearing. Earlier search code from the same overnight runs (local search, circulant search, the round-1 SAT stack) is untrusted by policy and cited nowhere in the argument. Confirm nothing in the chain silently depends on it.
7. **Literature claims.** Re-check the history of Section 1 against the primary sources, in particular the exact statement of Kaneko–Locke and the status of the 2026 minimum out-degree 7 preprint, and re-run the search for any published exhaustive small-order verification we may have missed.
8. **The $n = 15, 16$ upgrades.** If the two large proof verifications complete with `s VERIFIED` receipts, the headline can move to $n \leq 16$; that decision, and the wording, should be a human’s.

Reproduction appendix

All commands assume macOS with Homebrew; everything is CPU-only. On a shared machine, prefix heavy runs with `nice -n 15`.

1. Toolchain:

```
brew install cadical nauty python
pip install python-sat
git clone https://github.com/marijnheule/drat-trim && make -C drat-trim
```

2. Whole-space instances (assumption-free), $n = 6, \dots, 9$:

```
python3 encode.py N --out nN.cnf
cadical -q nN.cnf nN.drat # expect: s UNSATISFIABLE
drat-trim/drat-trim nN.cnf nN.drat # expect: s VERIFIED
```

3. Shard instances, $n = 10, \dots, 18$, $d = 1, \dots, \lfloor (n-1)/2 \rfloor$:

```
python3 encode.py N --mindeg D --exactdeg0 D --pin-nbhd \
  --sb-mode twoblock --out nN_sD.cnf
cadical -q nN_sD.cnf nN_sD.drat
drat-trim/drat-trim nN_sD.cnf nN_sD.drat
```

4. Whole-space cross-checks at $n = 10, 11$: `--mindeg 1 --sb-mode full` (and `--mindeg 3` for the lemma-conditioned variants).
5. Controls: `--slack` variants must come back SAT; truncating any `.drat` must make drat-trim report `s NOT VERIFIED`.
6. Pipeline B ($n \leq 8$; shard $n = 8$ via `res/mod`):

```
cc -O2 -o check_d6 check_d6.c
geng -q N | directg -o -q | ./check_d6
geng -q 8 r/10 | directg -o -q | ./check_d6 # r = 0..9, sum counts
```

7. Regenerate Table 1 from the artifacts, and re-verify every stored proof:

```
python3 gen_tables.py
sh verify_all_drat.sh
```

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